

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	24 July 2026
Team ID	Individual / Solo Project
Project Name	Software Installation Request Automation
Maximum Marks	4 Marks

Technical Architecture:

The diagram below shows the technical architecture for the Software Installation Request Automation solution, including application logic, infrastructure demarcation, external interfaces, data storage, and the machine learning component, followed by the component and characteristics tables.

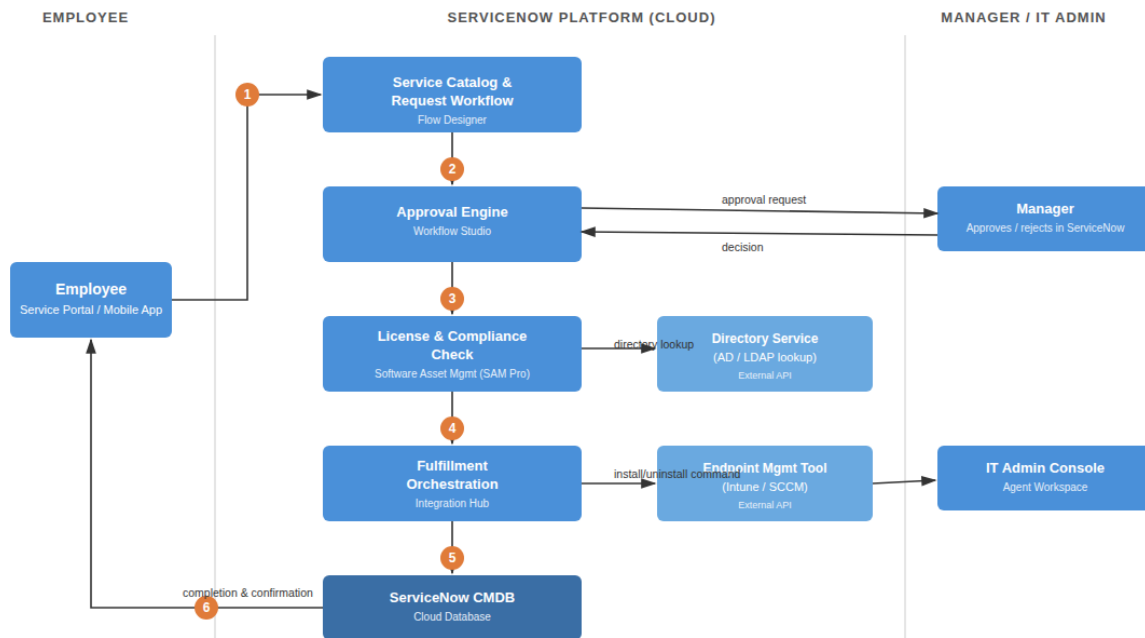


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How the employee interacts with the application	ServiceNow Service Portal, Now Mobile App
2	Application Logic-1	Service Catalog item and request intake workflow	ServiceNow Flow Designer / Workflow Studio
3	Application Logic-2	Manager approval routing and decisioning	ServiceNow Approval Engine
4	Application Logic-3	License availability and compliance check before fulfillment	Software Asset Management (SAM Pro)
5	Database	Core platform data — requests, approvals, configuration items	ServiceNow Platform Database (MySQL-based)
6	Cloud Database	Configuration Management Database, hosted on the ServiceNow cloud instance	ServiceNow CMDB
7	File Storage	Storage for attachments on requests (e.g. justification documents)	ServiceNow Attachment API
8	External API-1	Look up employee and manager details for approval routing	Active Directory / LDAP API

S.No	Component	Description	Technology
9	External API-2	Push and track remote software installation on the employee's device	Microsoft Intune / SCCM API
10	Machine Learning Model	Categorize and route incoming requests, and suggest similar past resolutions	ServiceNow Predictive Intelligence
11	Infrastructure (Server / Cloud)	Application deployment Local Server Configuration: MID Server for on-prem/AD connectivity Cloud Server Configuration: Multi-instance ServiceNow SaaS (Now Platform)	ServiceNow Cloud, MID Server

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used to build custom Service Portal widgets and UI pages	AngularJS, Bootstrap
2	Security Implementations	Role-based access control, single sign-on, and encryption for requests and license data	ACLs, SAML/SSO, OAuth 2.0, TLS encryption in transit, AES-256 at rest
3	Scalable Architecture	Multi-instance SaaS architecture separates catalog, workflow, and integration layers so each can scale independently	ServiceNow Now Platform (multi-instance SaaS)
4	Availability	Vendor-managed high availability across the ServiceNow cloud, with failover instances	ServiceNow HA/DR, load-balanced multi-instance
5	Performance	Caching of catalog items and asynchronous flow execution keep request submission and approval fast under load	Flow Designer async execution, CMDB indexing, catalog caching

References:

<https://c4model.com/>
<https://docs.servicenow.com/bundle/reference-pages/product/reference-pages>
<https://www.servicenow.com/products/software-asset-management.html>
<https://www.ibm.com/cloud/architecture>
<https://aws.amazon.com/architecture>